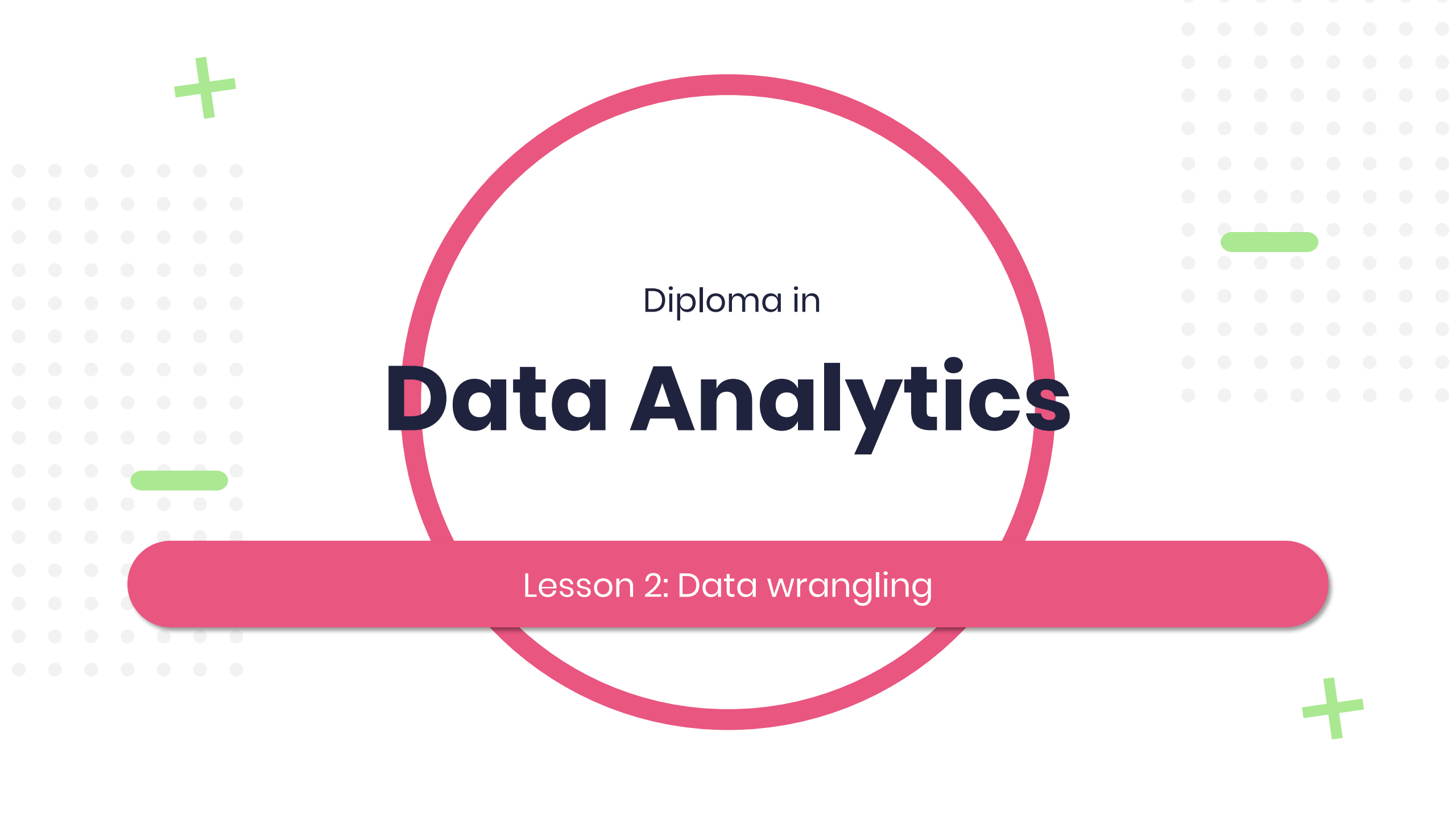
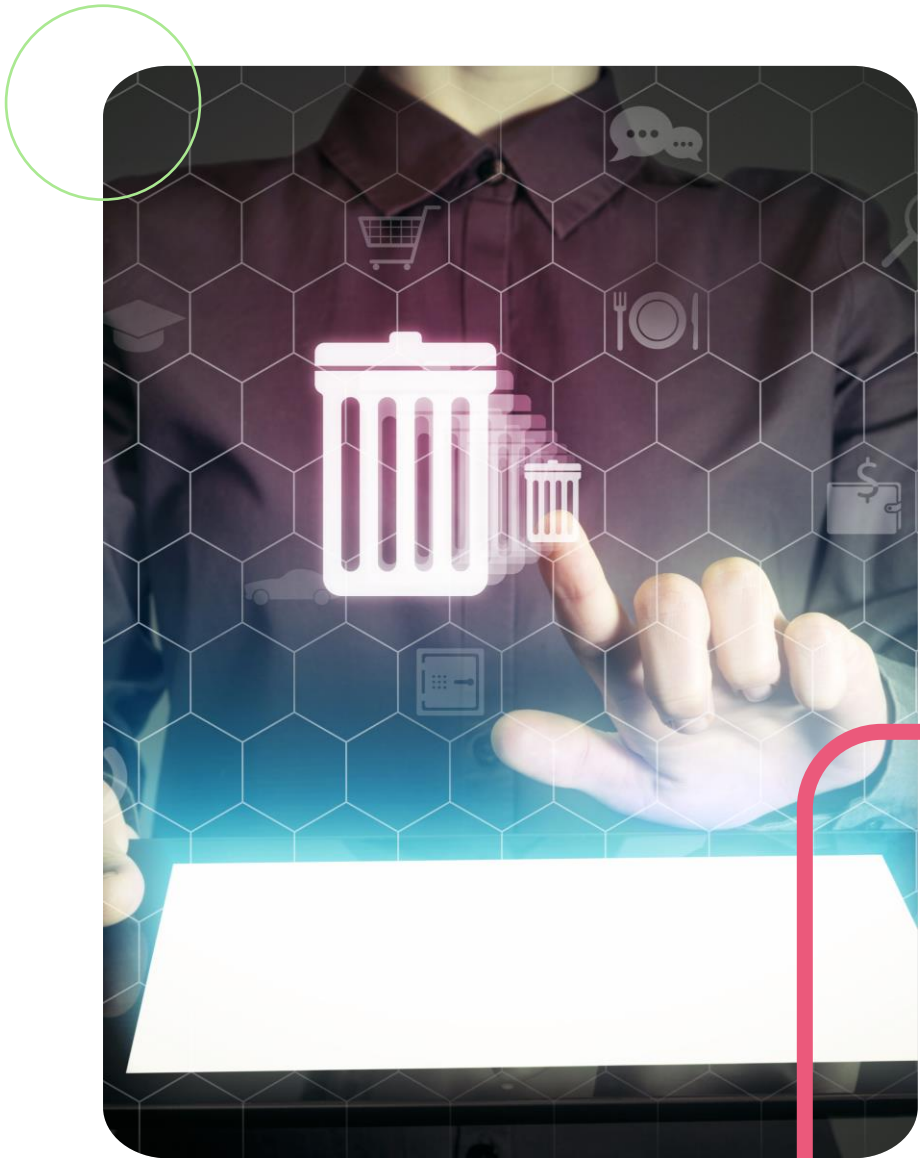




Diploma in

Data Analytics

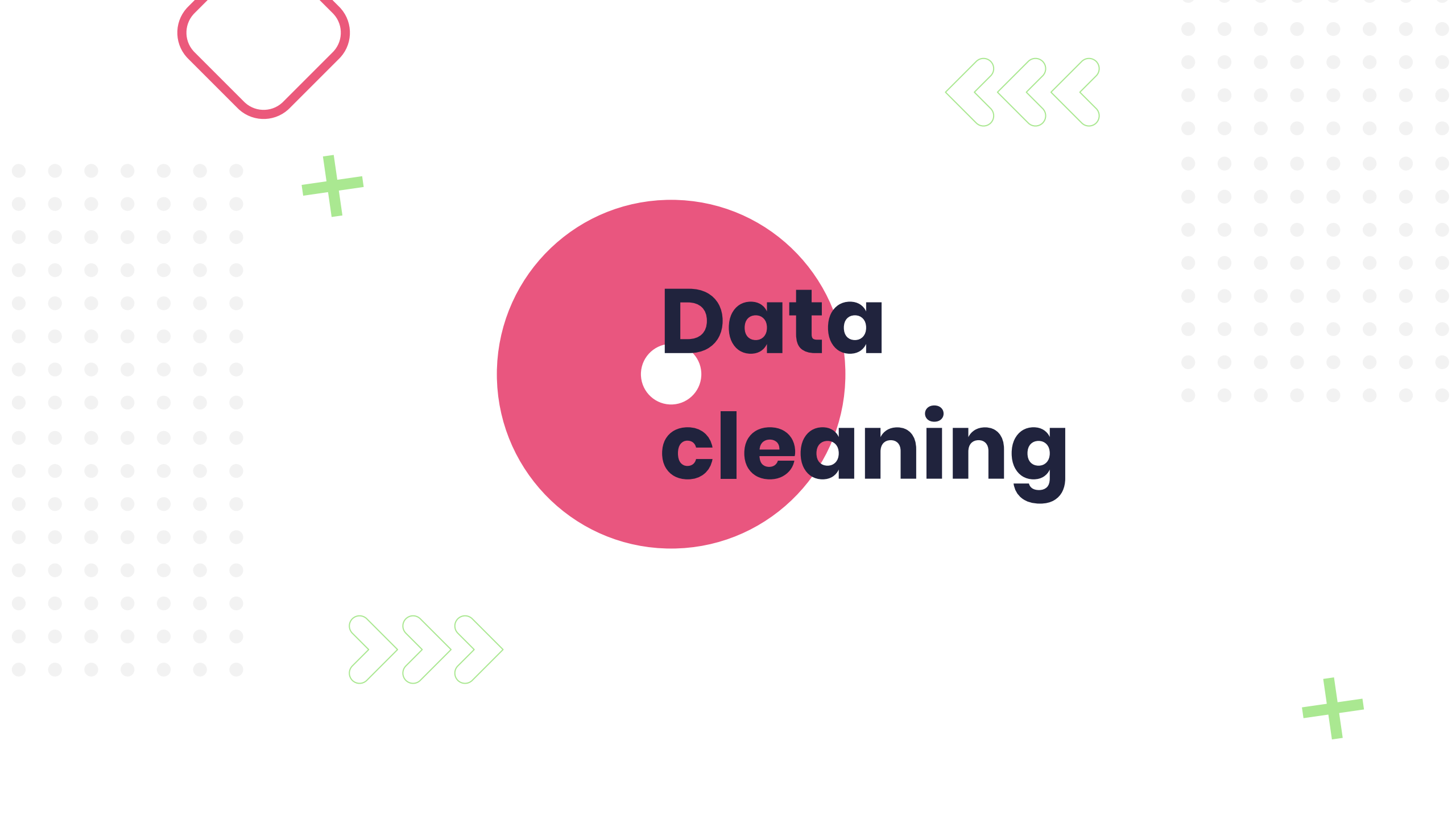
Lesson 2: Data wrangling



- Data cleaning <
- Described in R <
- Merging datasets <

Lesson Objectives

Lesson 2



Data cleaning

Data cleaning

- Process of detecting and correcting inaccuracies from data
- Up to 80% of the time



Recap packages in R

Collection of R functions,
compiled code and sample
data



Info on packages

- Description file
- Package Description (“package”)
- Help (package = “package”)





Hadley Wickham

What is tidy data?

“Tidy datasets are easy to manipulate, model and visualise, and have a specific structure: each variable is a column, each observation is a row, and each type of observational unit is a table.”



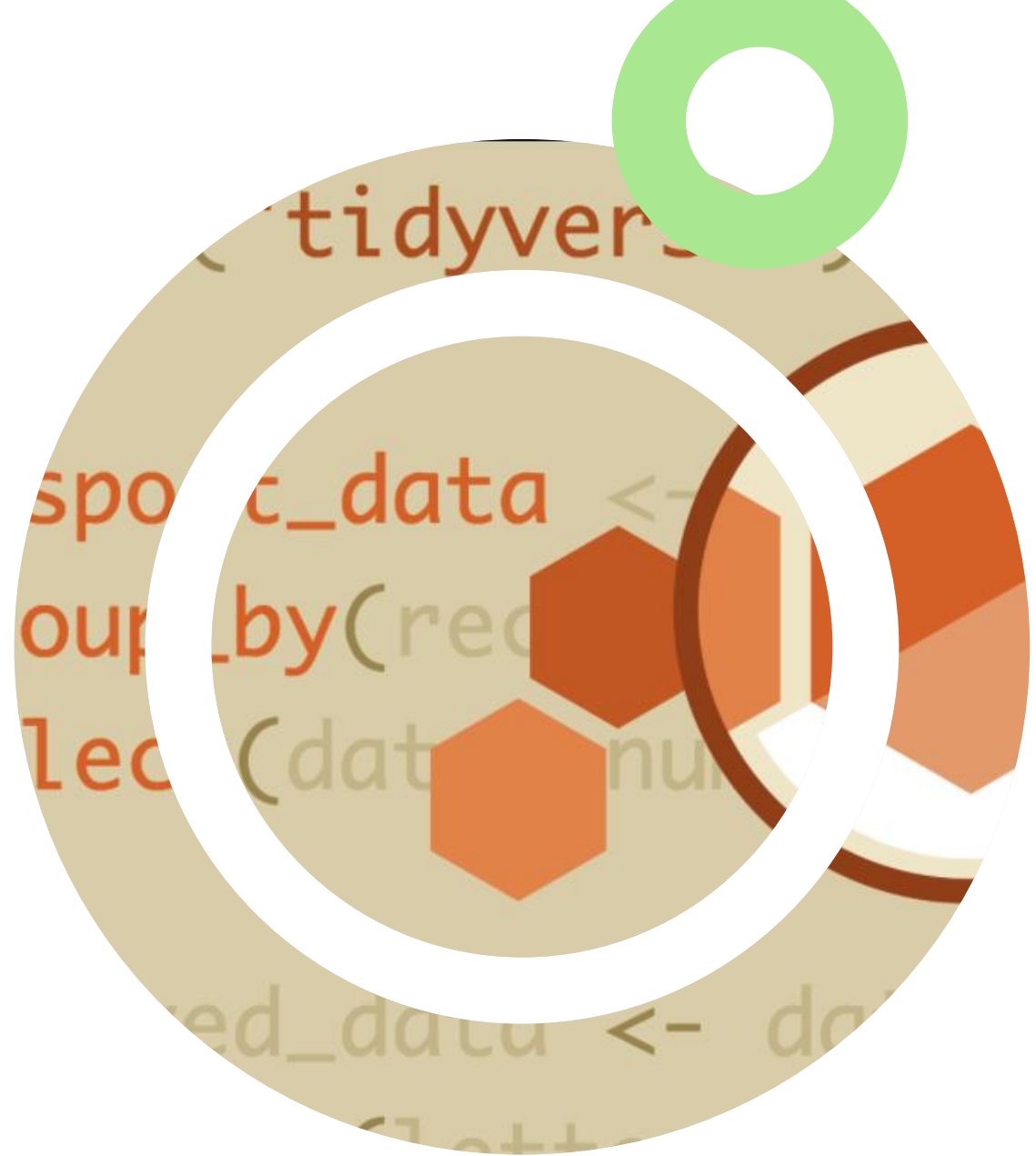
Why tidy data?

Tidy data is easier to manipulate, model and visualise.



What is tidyverse?

Collection of data analytics tools contained in R for transforming and visualising data



Packages available in tidyverse library

Ggplot2, dplyr, tidyr,
readr, purr, tibble,
stringr, forcats

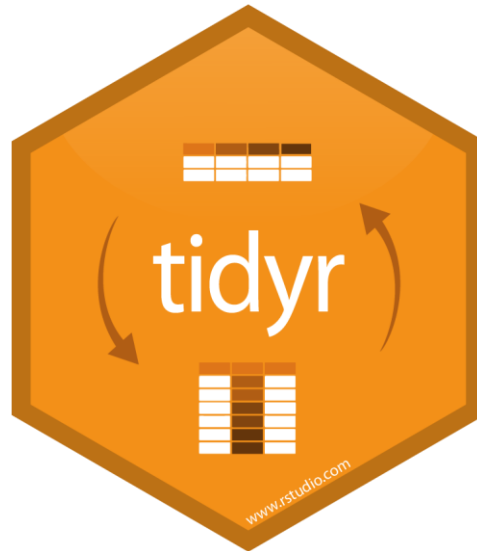


More about...

(packages contained in the tidyverse library)

tidyr

Helps create tidy data



dplyr

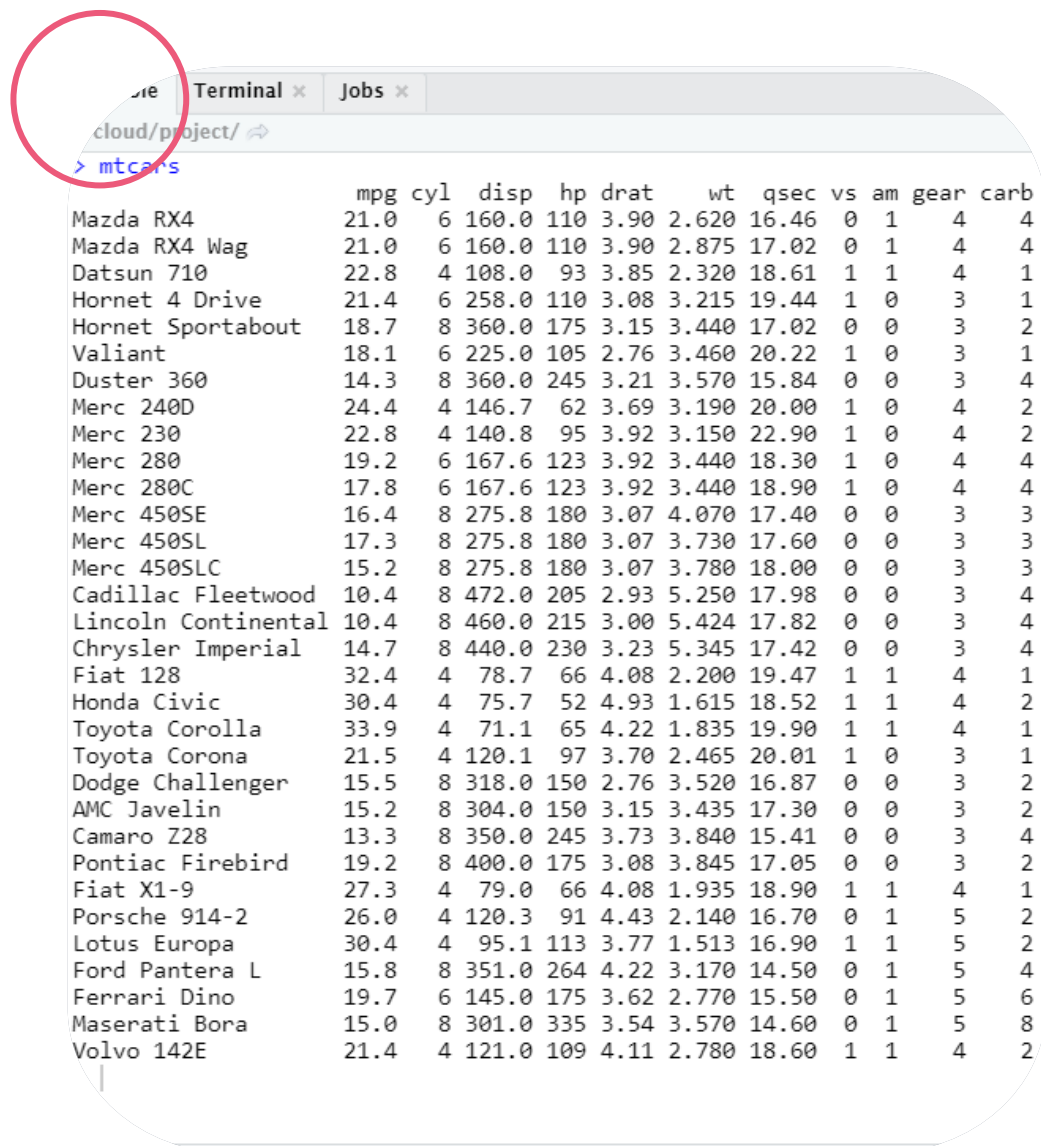
Helps manipulate data



readr

Helps import data





```

Terminal x Jobs x
cloud/project/ ↗
> mtcars

```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	6	225.0	105	2.76	3.460	20.22	1	0	3	1
Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0	0	3	4
Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	0	4	2
Merc 230	22.8	4	140.8	95	3.92	3.150	22.90	1	0	4	2
Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	0	4	4
Merc 280C	17.8	6	167.6	123	3.92	3.440	18.90	1	0	4	4
Merc 450SE	16.4	8	275.8	180	3.07	4.070	17.40	0	0	3	3
Merc 450SL	17.3	8	275.8	180	3.07	3.730	17.60	0	0	3	3
Merc 450SLC	15.2	8	275.8	180	3.07	3.780	18.00	0	0	3	3
Cadillac Fleetwood	10.4	8	472.0	205	2.93	5.250	17.98	0	0	3	4
Lincoln Continental	10.4	8	460.0	215	3.00	5.424	17.82	0	0	3	4
Chrysler Imperial	14.7	8	440.0	230	3.23	5.345	17.42	0	0	3	4
Fiat 128	32.4	4	78.7	66	4.08	2.200	19.47	1	1	4	1
Honda Civic	30.4	4	75.7	52	4.93	1.615	18.52	1	1	4	2
Toyota Corolla	33.9	4	71.1	65	4.22	1.835	19.90	1	1	4	1
Toyota Corona	21.5	4	120.1	97	3.70	2.465	20.01	1	0	3	1
Dodge Challenger	15.5	8	318.0	150	2.76	3.520	16.87	0	0	3	2
AMC Javelin	15.2	8	304.0	150	3.15	3.435	17.30	0	0	3	2
Camaro Z28	13.3	8	350.0	245	3.73	3.840	15.41	0	0	3	4
Pontiac Firebird	19.2	8	400.0	175	3.08	3.845	17.05	0	0	3	2
Fiat X1-9	27.3	4	79.0	66	4.08	1.935	18.90	1	1	4	1
Porsche 914-2	26.0	4	120.3	91	4.43	2.140	16.70	0	1	5	2
Lotus Europa	30.4	4	95.1	113	3.77	1.513	16.90	1	1	5	2
Ford Pantera L	15.8	8	351.0	264	4.22	3.170	14.50	0	1	5	4
Ferrari Dino	19.7	6	145.0	175	3.62	2.770	15.50	0	1	5	6
Maserati Bora	15.0	8	301.0	335	3.54	3.570	14.60	0	1	5	8
Volvo 142E	21.4	4	121.0	109	4.11	2.780	18.60	1	1	4	2

Data Frames

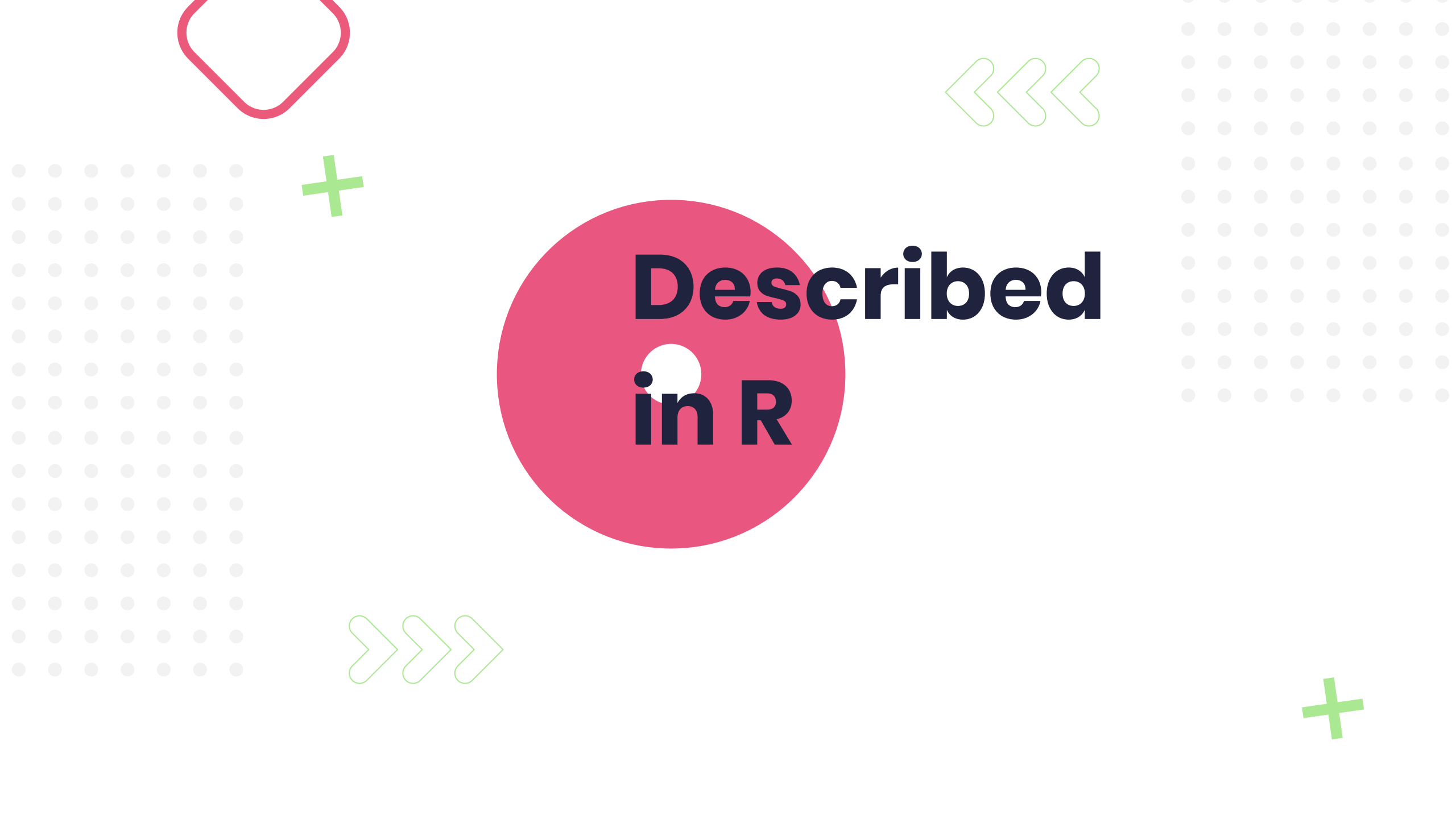
data.frame()

- R mostly works with data in data frames
- Table with rows and columns
- Similar to spreadsheet, table in SQL database

Titanic

Data management in Module 1
with Power Query





Described in R

Titanic

Described in Module 1 with Data
Analysis Toolpak

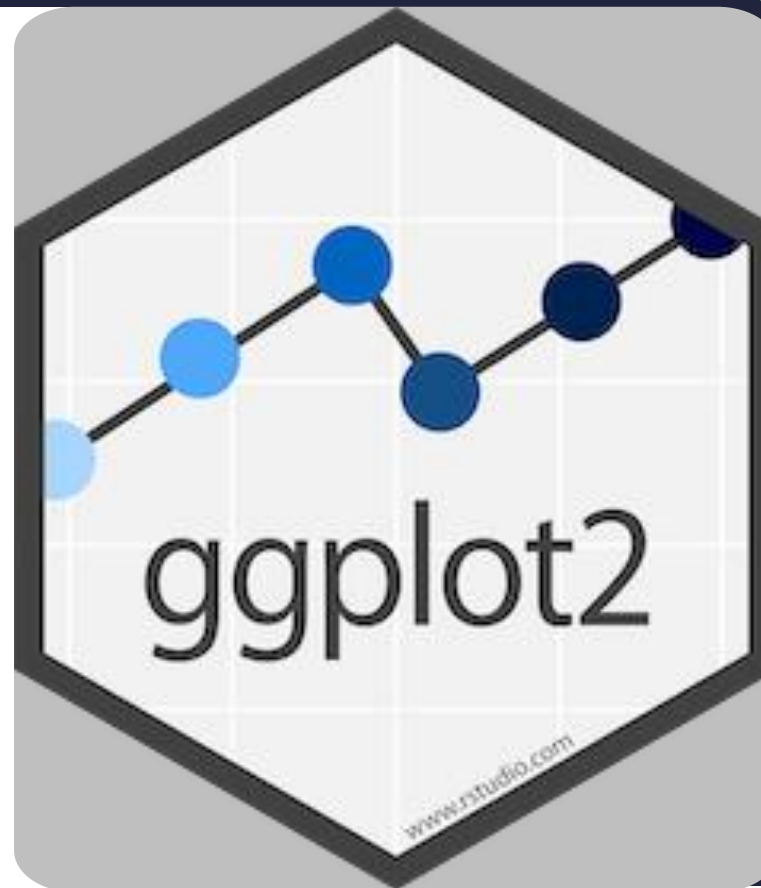


ggplot2

Helps create visualisations

Box -, density - & time series plots

Part of tidyverse



Help with R

RStudio community
Stack Overflow

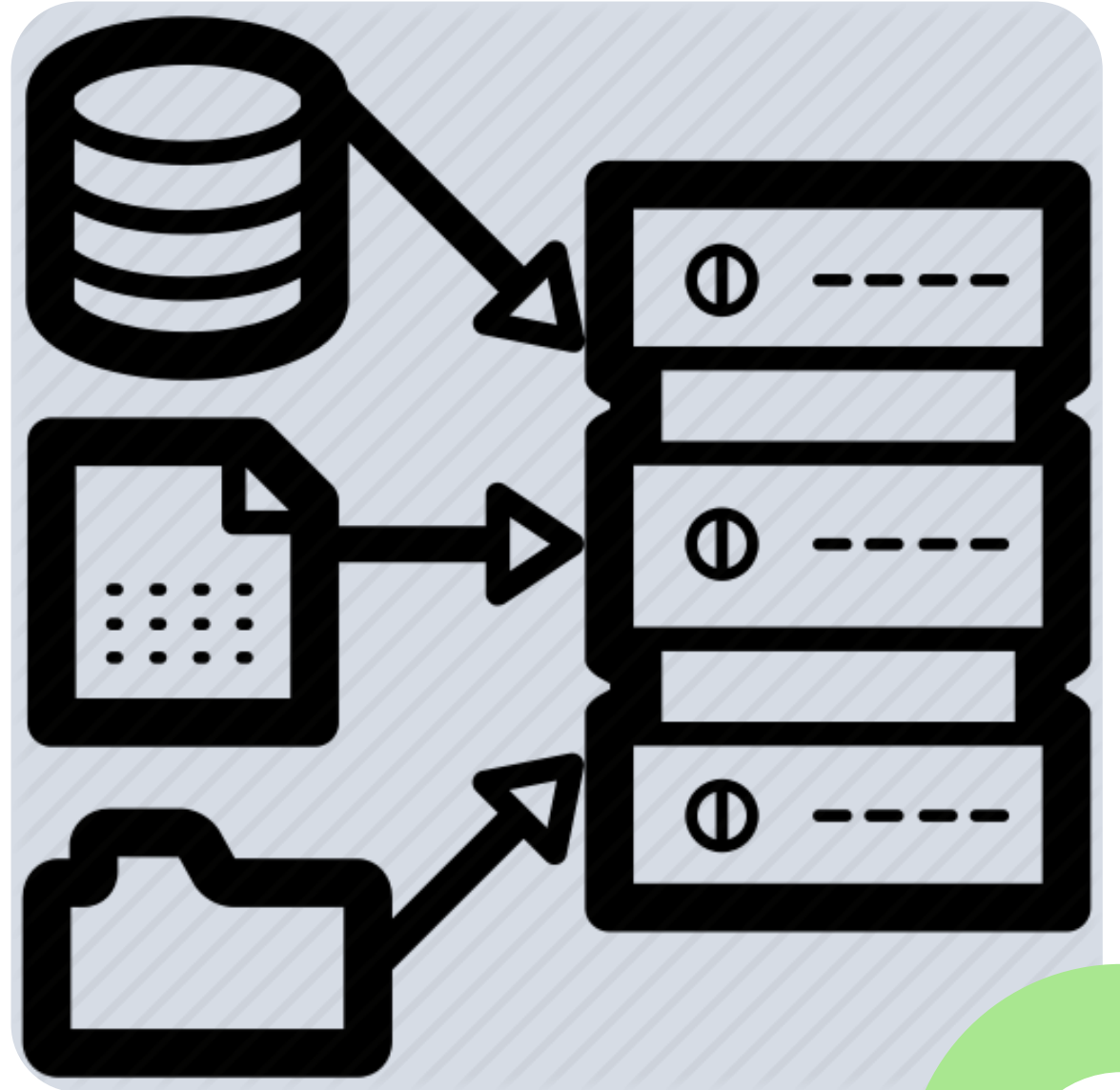


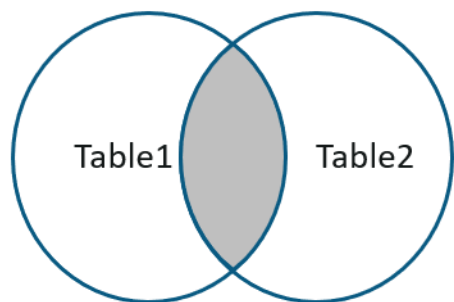


Merging datasets

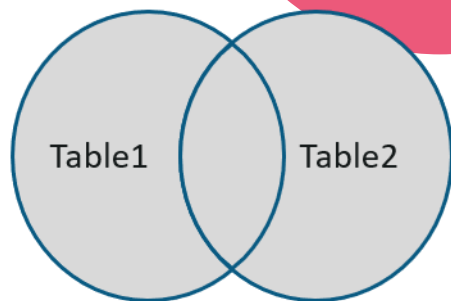
Merging datasets

- Bringing datasets together
- Combining datasets

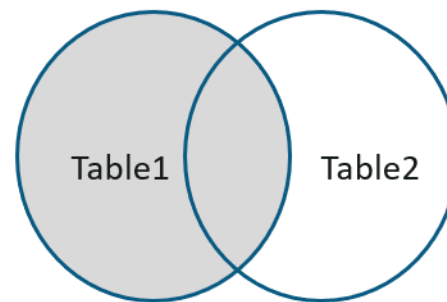




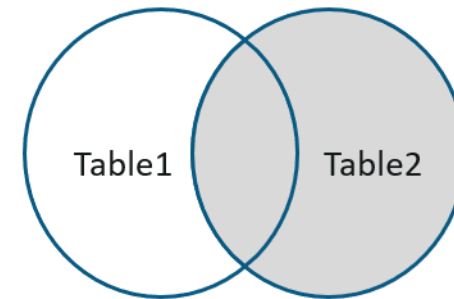
INNER JOIN



FULL JOIN



LEFT JOIN



RIGHT JOIN

dplyr for data manipulation

- `select()`
- `filter()`
- `group_by()`
- `summarise()`
- `arrange()`
- `join()`
- `mutate()`



Challenge >>

Join the training and test dataset from the Titanic and remove missing values from Age

#exploredata